NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR306487
Lead Applicant	Ms Amanda Male
Research Title	Finding out what causes dizziness that is triggered when changing position, to speed up access to the right treatment
Plain English Summary	**Title** Finding out what causes dizziness that is triggered when changing position, to speed up access to the right treatment **Background to the research** Dizziness affects 1-in-3 people at some stage in their lives. The main type is dizziness when changing position, such as getting into bed and rolling over. It can have a huge impact on a person's ability to do everyday activities, limiting how much they move or walk and can cause falls. To get help, people visit doctors and physiotherapists in a range of healthcare settings including general practitioner clinics, emergency departments, physiotherapy services, falls services and specialist clinics. The main cause of dizziness when changing position, is due to small crystals getting lodged in a part of the inner ear where they are not meant to be. When the person moves they experience dizziness due to the crystal moving. Doctors and physiotherapists can relocate these crystals, by moving the person through a series of specific positions. Sometimes, dizziness when changing position can be caused by brain problems but many doctors and physiotherapists are less aware of this. There are several problems doctors and physiotherapists face when treating this type of dizziness, specifically: • Many are unaware that conditions affecting the brain can also cause this type of dizziness and these conditions need to be treated differently • We do not know how many treatments are needed to cure an inner ear problem and how often it returns • We do not know why some people do not get better **Aims of the research** Aim 1 - develop a list of the clinical features that suggest a brain cause of the dizziness Aim 2 - identify how many treatments are needed and how often it comes back for an inner ear cause of the dizziness Aim 3 - explore if there are differences in the inner ear anatomy for those that do not get better with the specific treatment positions **Design and methods used** Study 1 – I will complete a thorough review of all the research on brain causes of dizziness with position changes. This information will be combined with the findings from study 2 to create a list of features to identify brain causes of dizziness. Study 2 – I will invite 250 people who have dizziness when they change position to take part. They will fill in questionnaires, have a thorough dizziness assessment, and treatment when appropriate. Information on the number of treatments required, and recurrence will be collected. Study 3 – I will invite 25 people that get better with treatment, and 25 people that do not get better, to have scans of their inner ears. 3D models will be made from the scans and compared for any differences in the anatomy of the inner ears. **Patient and public involvement** Ten people with experience of this type of dizziness, expressed frustrations in getting a diagnosis quickly, and access to the right treatment. Their feedback has shaped the research question and study design, specifically: • How to advertise the study • Printing materials, instead of relying on email alone • Providing interpreters • Refunding travel costs Another group of eight patients have agreed to help with recruitment and share the results widely. **Dissemination** Doctors and physiotherapists are the key audience to share these results with, to raise awareness of the causes and treatment of dizziness caused by changing position, to improve diagnosis and treatment. This will be done through publications in healthcare journals, presentations at international conferences and courses.

	The results will also be shared with patients and public using social media and an infographic, newsletters and at a specific workshop.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

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1. Application Summary Information

Host organisation (which will administer any award)
University College London

Partner organisation (if applicable)	University College London Hospitals NHS Foundation Trust

Application Type

Strategic Theme	N/A

Partnership Fellowship	N/A

WTE % Please select which WTE option you would prefer
1.00

Duration Doctoral Fellowship is a 3 year award and based on your WTE preference the duration in months is calculated as follows
36

Research Title	Finding out what causes dizziness that is triggered when changing position, to speed up access to the right treatment

Research Type	Other (please specify)

Please provide more details on your selection above.
Prospective longitudinal observational study

Proposed start date if grant awarded	01/01/2026

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

The 'Publication Record' section of the form is automatically populated from publications added into the "My Research Outputs" page of your ARAMIS account.

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class	Department
01/2022	04/2022	International Online Vestibular Diploma	Diagnosing and Managing Vestibular Disorders: A Knowledge-based Competency Course	United Kingdom (England)	Audiology-Vestibular International Science Academy (AVISA)	91%	N/A
09/2014	09/2015	MResCP	Master of Research in Clinical Practice (NIHR funded)	United Kingdom (England)	St George's University of London	70% Merit	Faculty of Health, Social Care and Education
01/2011	03/2011	Master module	Cognitive and Behavioural Aspects of Neurorehabilitation	United Kingdom (England)	Brunel University of London	C	Health Science
09/2010	12/2010	Master module	Functional Neuroscience for Rehabilitation	United Kingdom (England)	Brunel University of London	A	Health Science
09/1999	06/2002	BSc(Hons)	Physiotherapy	United Kingdom (England)	Sheffield Hallam University	2:1	Health and Social Care

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
12/2021	-	Honorary Research Associate (unpaid)	SENSE Research Unit, Department of Clinical & Movements Neurosciences, UCL Institute of Neurology	20
12/2021	-	Highly Specialised Vestibular Physiotherapist	Vestibular Rehab Physio	0
04/2011	12/2021	Highly Specialised Vestibular Physiotherapist	Therapy Services The National Hospital for Neurology and Neurosurgery, London	0
11/2007	04/2011	Senior I (Band 7) physiotherapist	Neurological Rehabilitation Unit The Wellington Hospital, London	20
01/2007	10/2007	Senior II (Band 6) physiotherapist	Neurological Rehabilitation Unit The Wellington Hospital, London	0
12/2003	12/2006	Senior II (Band 6) physiotherapist - Neurology	Therapy Services Heatherwood and Wexham Park Hospitals NHS Trust	0

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
07/2002	12/2003	Junior (Band 5) Rotational physiotherapist	Therapy Services Heatherwood and Wexham Park Hospitals NHS Trust	0

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
Meniere's Society	Contribution to the first years costs of the research fellowship.	55106	Meniere's Society	12	01/01/2026	31/12/2026

Publication Record
(include the list of authors, article title, journal name, and volume)
This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Koohi, N., Holmes, S., **Male, A.**, Bamiou, DE., Dudziec, MM., Ramdharry, GM., Pizzamiglio, C., Hanna, MG., Pitceathly, RDS. and Kaski, D.. (2024) ‘Beyond the cochlea: exploring the multifaceted nature of hearing loss in primary mitochondrial diseases.’, *Brain communications*, 6(6).
Male, AJ., Koohi, N., Holmes, SL., Pitceathly, RDS. and Kaski, D.. (2024) ‘Acceptability of Audiovestibular Assessment in the Home-A Patient Survey.’, *Audiology research*, 14(3).
Burrows, L., Lesser, T. and **Male, A.**. (2024) ‘Developments in Dizziness and Balance Disorders.’, *The Journal of laryngology and otology*, 138(S2).
Male, AJ., Holmes, SL., Koohi, N., Dudziec, M., Hanna, MG., Ramdharry, GM., Pizzamiglio, C., Pitceathly, RDS. and Kaski, D.. (2024) ‘A diagnostic framework to identify vestibular involvement in multi-sensory neurological disease.’, *European Journal of Neurology*.
Koohi, N., **Male, AJ.** and Kaski, D.. (2023) ‘Acute positional vertigo in the emergency department-peripheral vs. central positional nystagmus.’, *Frontiers in neurology*, 14.
Male, AJ., Holmes, SL., Hanna, MG., Pitceathly, RDS., Ramdharry, GM. and Kaski, D.. (2023) ‘Correction to: Development of a diagnostic framework for vestibular causes of dizziness and unsteadiness in patients with multisensory neurological disease: a Delphi consensus.’, *Journal of neurology*, 270(8).
Male, AJ., Holmes, SL., Hanna, MG., Pitceathly, RDS., Ramdharry, GM. and Kaski, D.. (2023) ‘Development of a diagnostic framework for vestibular causes of dizziness and unsteadiness in patients with multisensory neurological disease: a Delphi consensus.’, *Journal of neurology*, 270(6).
Stancel-Lewis, J., Lau, JWL., **Male, A.**, Korres, G., Rogel-Salazar, J., Pavlou, M. and Bamiou, DE.. (2022) ‘Vestibular Rehabilitation Therapy for the Treatment of Vestibular Migraine, and the Impact of Traumatic Brain Injury on Outcome: A Retrospective Study.’, *Otology & neurotology : official publication of the American Otological Society, American Neurotology Society [and] European Academy of Otology and Neurotology*, 43(3).
Bennett, E., Holmes, S., Koohi, N., Islam, S., Bancroft, M., **Male, A.**, Hanna, MG., Pitceathly, RDS. and Kaski, D.. (2022) ‘Self-reported postural symptoms predict vestibular dysfunction and falls in patients with multi-sensory impairment.’, *Journal of neurology*, 269(5).
Tahtis, V., **Male, A.** and Kaski, D.. (2021) ‘Positional Manoeuvres for BPPV: Theoretical Approach to Remote Training for Non-specialists.’, *Frontiers in neurology*, 12.
Holmes, S., **Male, AJ.**, Ramdharry, G., Woodward, C., James, N., Skorupinska, I., Skorupinska, M., Germain, L., Kozyra, D., Bugiardini, E., Poole, OV., Quinlivan, R., Hanna, MG., Kaski, D. and Pitceathly, RDS.. (2018) ‘Vestibular dysfunction: a frequent problem for adults with mitochondrial disease.’, *Journal of neurology, neurosurgery, and psychiatry*, 90(7).
Male, AJ., Ramdharry, GM., Grant, R., Davies, RA. and Beith, ID.. (2018) ‘A survey of current management of Benign Paroxysmal Positional Vertigo (BPPV) by physiotherapists' interested in vestibular rehabilitation in the

Publication Record

(include the list of authors, article title, journal name, and volume)
This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
UK.’, *Physiotherapy*, 105(3).
Kaski, D., Haider, S., **Male, A.**, Radunovich, A., Liu, F., Cordivari, C., Bhatia, KP. and Bronstein, AM.. (2018) ‘Adult Periodic Alternating Nystagmus Masked by Involuntary Head Movements.’, *Frontiers in neurology*, 9.

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/05/2019	The Lily Foundation	£84,742 provided to develop and validate an assessment framework to improve the diagnosis and management of vestibular dysfunction in mitochondrial disease.	Co-applicant and co-lead for the project
01/09/2014	National Institute for Health and Care Research	Full time funded award to undertake a Masters in Research and Clinical Practice at St George's University of London.	Applicant

ORCID iD 0000-0001-5508-908X

3. Applicant Research Background

Applicant Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Physiotherapist

Please describe your research career to date

I am recognised as one of the clinical leaders in the field of vestibular physiotherapy in the United Kingdom. I believe I have the experience, ambition and motivation to successfully complete this doctoral fellowship and become a future clinical academic leader in the specialty.

Research experience

I began my research journey during a clinical physiotherapy position, where I collaborated with a colleague to conceptualise, design, and conduct a pilot study on a novel intervention for upper limb function in the sub-acute stroke population (January 2009–January 2011). This quantitative study, involving five patients, was presented as a poster at the World Confederation for Physiotherapy (June 2011). The experience provided valuable insights into the research process and sparked my passion for integrating research with clinical practice.

I pursued a full-time Master's of Research in Clinical Practice at St. George's University of London (September 2015), funded by the NIHR. This program honed my skills in designing, executing, analysing, and disseminating research using quantitative and qualitative methodologies. My dissertation involved a mixed-methods survey of physiotherapists' management of benign paroxysmal positional vertigo (BPPV) in the UK, which required university ethics approval. The findings were published in the Journal of Physiotherapy (September 2019) and presented as posters at the European Region of World Confederation for Physiotherapy (November 2021).

Returning to my NHS role, I co-led a project identifying vestibular dysfunction in mitochondrial disease patients, in the largest cohort studied to date. This research was published in the Journal of Neurology, Neurosurgery and Psychiatry (November 2018) and presented at the Bárány Society Meeting (June 2018). Building on these results, we secured funding from The Lily Foundation (May 2019) for a prospective observational study. I co-developed a diagnostic framework for vestibular dysfunction in mitochondrial disease, supervised research associates, managed data, and adapted the methodology during the COVID-19 pandemic. I created an electronic case record form and coordinated research activities within the CPMS. I conducted a modified Delphi consensus for the framework development, published in Journal of Neurology (June 2023) and shared as a poster at the Bárány Society Meeting (May 2022). I led the dissemination of the framework's sensitivity and specificity results in the European Journal of Neurology (May 2024) and published a patient survey analysis in Audiology Research (June 2024).

In collaboration with the acute vertigo service, I co-authored a publication highlighting the prevalence of central versus peripheral causes of positional vertigo and common diagnostic errors by non-specialists. These findings were presented at the British Society of Neuro-Otology annual conference (October 2023).

My research endeavours have fuelled my interest in positional dizziness and equipped me with robust skills in designing, conducting, and disseminating studies. Beyond my projects, I have peer-reviewed manuscripts for various journals and co-edited a specialist supplement on dizziness and balance disorders for the Journal of Laryngology and Otology (May 2024). I also co-authored a British Society of Audiology practice guideline on vestibular rehabilitation (April 2019) and contributed two case studies to the book "*Puzzling Cases in Vertigo and Dizziness*".

Clinical and leadership experience

I have worked clinically as a vestibular and neurological physiotherapist since 2002, and have developed and led three vestibular rehabilitation services, within secondary and tertiary NHS services and the private sector. I have developed advanced clinical skills in vestibular assessment and treatment, through mentoring by medical and vestibular specialists, independent professional development, and attendance at specialist courses in the UK and overseas.

I was responsible for the supervision of junior colleagues, appraisals, and performance management. I have supported team members to analyse data, present audits, develop service improvement ideas and report the findings to managers. I have helped others design posters for conferences and pass these skills onto more junior team members. In my last position, I supported the entire therapy outpatient department in a major transformation project to adopt an online documentation system and coordinated two department relocations and changes in service delivery.

My interest in vestibular rehabilitation as a speciality, led me to become one of the founding members of the Chartered Society of Physiotherapy special interest group – the Association of Chartered Physiotherapist interested in Vestibular Rehabilitation (ACPIVR). I was a committee member for eight years (2008-2016), culminating in 2 years as vice chairperson, and the final two years as chairperson. This committee was responsible for developing the knowledge and skills of other physiotherapists nationally, raising awareness of the specialism, and developing resources for clinicians and patients.

I am passionate about providing evidence-based treatment and support others to do the same. I frequently provide training locally, nationally and internationally, to physiotherapists and a variety of medical professionals on vestibular rehabilitation and BPPV management. I lecture on several University College London (UCL) Master's programs and have supervised three Master's student's research projects, to include supporting one to publish their findings in a peer reviewed journal.

Impact of Fellowship

I am privileged to have an honorary position with one of the world's leading neuro-otology research units at UCL, a testament to my standing as one of the UK's most experienced and respected vestibular physiotherapists. My strong relationships within the multidisciplinary clinical and research community have been instrumental in advancing patient care and contributing to impactful research.

Securing this fellowship will empower me to pursue research on a topic I am deeply passionate about, whilst addressing the most common cause of dizziness worldwide. This work has the potential to significantly enhance patients' quality of life while alleviating the strain on overburdened NHS services. The fellowship will provide essential resources, dedicated time, and access to leading research experts, ensuring the project is not only feasible but also methodologically rigorous.

Towards the end of this fellowship, I hope to establish a part time NHS position at UCLH. This will allow the new referral pathway and service to continue upon study completion, whilst I apply for an NIHR advanced clinical and practitioner academic fellowship. This fellowship therefore represents a crucial step in realising my vision to drive meaningful change in vestibular health on both national and global scales and advance my academic career.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details
This research project was submitted to the Meniere's Society, the UK's balance and dizziness charity, to request research funding. The application was peer reviewed by several external experts and layperson committee members, and feedback was given to the trustees. A decision was given in December 2024 to award £55,106.04 towards the project. The award letter stated they felt the project was "…important and would be very beneficial for patients with dizziness and unsteadiness". Unfortunately, as they are a small charity with limited funds and receive numerous grant applications each year, they were unable to award the full amount requested. However, they are very enthusiastic about a joint collaboration with the NIHR and fully support this research fellowship application. They are keen to help throughout the project including with recruitment and dissemination via presentations, their newsletter and at their balance awareness week events.

This fellowship application reflects the total cost of £447,139.00 that is required to undertake this research project, remunerate patient and public involvement, undertake relevant professional and academic training, attend conferences, and complete a three year full-time PhD. The amount awarded by the Meniere's Society of £55,106.04, will be deducted from the total cost, should this application be successful.

This application has also been reviewed by the NIHR research support service hub and underwent a grantsmanship review. All feedback has been taken onboard and addressed and I believe this has strengthened the application.

Contextual Factors

Please use this section to detail any factors that may have impacted the development of your research career or outputs from your research career to date. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research activities (including research outputs), due to disability or to caring responsibilities. This may include physical or mental health related impairments or conditions, which have impacted your opportunities to undertake research activities
- Reduced opportunities for career support e.g., mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, contextual factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with the General Data and Protection Regulations (GDPR).

The last NHS position I held had no research time apportioned and the data collection phase of the observational study we had received charity funding (Lily Foundation) for, had finished. There was a large amount of data that required analysing, findings to share with the patients and public, and professional manuscripts to be written.

I am committed to progressing my career as a clinical academic researcher, and my obligations to disseminate results from my research activities. As such, I made the decision to leave my full-time NHS position to allow time to focus on my research and took up a part time, honorary (unpaid) contract role based in the SENSE research unit, at UCL. This part time role has allowed the time to prepare and successfully publish three articles relating to the Lily Foundation research grant and assist in one other. It also allowed time to publish one more article relating to this fellowship topic and be an editor for a standalone supplement in the Journal of laryngology and otology. Moreover, alongside the research team, I continue to analyse the data to address secondary variables.

In addition, this part time role in the SENSE research unit has provided a rich opportunity to collaborate and network with current UCL PhD students and other distinguished researchers. It has provided time to undertake the patient focus group, establish a patient advisory group for this project, and the time to develop this fellowship application to advance my academic career.

In addition to my professional ambitions, I have qualified for Great Britain's Age Group Triathlon teams, for both the European Championships (sprint distance) and World Championships (Olympic distance) in 2025. The last competition takes place in Australia in October 2025, and this is why I have deferred the start date of this research fellowship until January 2026 to allow these competitions to have concluded. I have the support of my supervisory team, to ensure I will be able to manage this research fellowship in future, alongside the training for triathlon events, so that I can pursue both my professional and sporting ambitions.

4. Plain English Summary of Research

Title

Finding out what causes dizziness that is triggered when changing position, to speed up access to the right treatment

Background to the research

Dizziness affects 1-in-3 people at some stage in their lives. The main type is dizziness when changing position, such as getting into bed and rolling over. It can have a huge impact on a person's ability to do everyday activities, limiting how much they move or walk and can cause falls.

To get help, people visit doctors and physiotherapists in a range of healthcare settings including general practitioner clinics, emergency departments, physiotherapy services, falls services and specialist clinics.

The main cause of dizziness when changing position, is due to small crystals getting lodged in a part of the inner ear where they are not meant to be. When the person moves they experience dizziness due to the crystal moving. Doctors and physiotherapists can relocate these crystals, by moving the person through a series of specific positions. Sometimes, dizziness when changing position can be caused by brain problems but many doctors and physiotherapists are less aware of this.

There are several problems doctors and physiotherapists face when treating this type of dizziness, specifically:

- Many are unaware that conditions affecting the brain can also cause this type of dizziness and these conditions need to be treated differently
- We do not know how many treatments are needed to cure an inner ear problem and how often it returns
- We do not know why some people do not get better

Aims of the research

Aim 1 - develop a list of the clinical features that suggest a brain cause of the dizziness

Aim 2 - identify how many treatments are needed and how often it comes back for an inner ear cause of the dizziness

Aim 3 - explore if there are differences in the inner ear anatomy for those that do not get better with the specific treatment positions

Design and methods used

Study 1 – I will complete a thorough review of all the research on brain causes of dizziness with position changes. This information will be combined with the findings from study 2 to create a list of features to identify brain causes of dizziness.

Study 2 – I will invite 250 people who have dizziness when they change position to take part. They will fill in questionnaires, have a thorough dizziness assessment, and treatment when appropriate. Information on the number of treatments required, and recurrence will be collected.

Study 3 – I will invite 25 people that get better with treatment, and 25 people that do not get better, to have scans of their inner ears. 3D models will be made from the scans and compared for any differences in the anatomy of the inner ears.

Patient and public involvement

Ten people with experience of this type of dizziness, expressed frustrations in getting a diagnosis quickly, and access to the right treatment. Their feedback has shaped the research question and study design, specifically:

- How to advertise the study
- Printing materials, instead of relying on email alone
- Providing interpreters
- Refunding travel costs

Another group of eight patients have agreed to help with recruitment and share the results widely.

Dissemination

Doctors and physiotherapists are the key audience to share these results with, to raise awareness of the causes and treatment of dizziness caused by changing position, to improve diagnosis and treatment. This will be done through publications in healthcare journals, presentations at international conferences and courses.

The results will also be shared with patients and public using social media and an infographic, newsletters and at a specific workshop.

5. Scientific Abstract

Background

Positional dizziness accounts for nearly 50% of dizziness cases seeking medical attention. It describes dizziness, vertigo and unsteadiness triggered by changes in head position, and heavily impacts patients' independence, mood, and falls risk.

Patients present across a range of healthcare settings, including general practitioner's clinics, emergency departments, physiotherapy services, falls services and specialist clinics.

Most cases are due to five subtypes of benign paroxysmal positional vertigo (BPPV), but brain (central) disorders such as tumours, stroke, or vestibular migraine are under-recognised causes of positional dizziness. It is important to determine the cause to ensure expedient management because the treatments provided by doctors and physiotherapists for BPPV will not work for central causes and they may require emergency treatment.

Patients in a focus group expressed frustrations at getting a diagnosis and access to prompt and effective treatment, reinforcing the major challenges faced by doctors and physiotherapists in this area, specifically:

- poor awareness of the diagnostic features of central causes of positional dizziness
- limited evidence on the number of treatments required to resolve each BPPV subtype and the recurrence rates for each BPPV subtype
- uncertainty regarding reasons for poor treatment response and whether an anatomical variation of the inner ear could explain this

This research intends to address these gaps in knowledge to refine diagnostic frameworks, improve access to timely effective treatment by doctors and physiotherapists, and improve patient quality of life.

Aims and objectives

- Review (i) the key diagnostic characteristics for central causes of positional dizziness and (ii) explore which differentiate central from peripheral dizziness
- Characterise the diagnostic features for peripheral and central causes of positional dizziness
- Determine how many treatments are required and the recurrence rates for each subtype of BPPV
- Explore possible anatomical differences to the inner ear affecting treatment response

Methods

Study 1 – Complete a systematic review to identify the reported diagnostic characteristics for central causes of positional dizziness, to incorporate into study 2.

Study 2 – Recruit 250 adults living in North Central London, reporting positional dizziness. Participants will complete dizziness related questionnaires and undergo a standardised assessment using a specialist treatment chair, wearing video-goggles to record and quantify eye movements. Data will be collected on BPPV treatments needed and recurrence rates.

Study 3 – Conduct a pilot study to explore if an anatomical variation accounts for non-response to BPPV treatment in study 2. Reconstruct three-dimensional models from magnetic resonance imaging scans of the inner ear, and compare 25 participants that respond to treatment, to 25 that do not.

Anticipated impact and dissemination

This research will impact doctors and physiotherapists managing patients with positional dizziness, but also patient wellbeing. Diagnostic features for central positional dizziness will be identified and existing knowledge gaps filled for BPPV, specifically the number of treatments needed, recurrence rates and why some patients do not get better. Findings will be published in medical journals and presented at conferences and courses, to inform diagnostic and treatment guidelines, and improve patient care. An infographic and dedicated workshop will help share the results with patients.

6. Detailed Research Plan

Title:

Finding out what causes dizziness that is triggered when changing position, to speed up access to the right treatment

This research fellowship aims to expand the limited literature to help doctors and physiotherapists identify and distinguish between inner ear and brain causes of positional dizziness more effectively, where treatment and outcomes differ substantially. The overall objective is to improve access to the correct diagnosis and treatment in a more timely manner and thus enhance patient care and experience.

1. What is the problem being addressed?

Dizziness ranks among the top reasons for visiting a doctor, affecting up to 35% of individuals at some point in their lives[1]. Dizziness that is triggered by a change in head position, for example, getting into bed, rolling in bed, and bending over, is the main cause[1,2]. Such positional dizziness can result in dizziness and vertigo[2], with unsteadiness commonly reported[3]. The symptoms are typically brief but can have a significant impact on independence in everyday activities[4], negatively impacting mood and walking confidence[4]. Of greatest concern in the elderly, positional dizziness is associated with high falls risk[4], with associated morbidity and mortality[4,5].

Patients with positional dizziness can present to doctors and physiotherapists working in a wide range of healthcare settings, including general practitioners clinics, emergency departments, physiotherapy services, falls services and specialist neuro-otology departments[3,4,6].

Most cases of positional dizziness are due to an inner ear condition called benign paroxysmal positional vertigo (BPPV), with the majority of occurrences easily cured[1]. However, a small proportion of cases are due to brain disorders, that if left undetected and untreated can have life-threatening consequences[1,7].

There is limited detailed prevalence data on positional dizziness across healthcare settings, with research failing to differentiate between inner ear (peripheral) and brain (central) causes or reporting on just one cause in the study's population. The most comprehensive data reports in 100 patients that had position triggered dizziness, 80% had BPPV and 20% had central causes[7].

One challenge related to positional dizziness is that it can be due to peripheral or central disorders[2]. Bedside, gold standard positioning manoeuvres are needed to make the correct diagnosis[3,8] but these tests rely on correct execution and interpretation of the induced eye movement (nystagmus). This can be challenging for doctors or physiotherapists that have little or no training in managing dizziness[9] where these patients may present.

It is important to identify whether positional dizziness is due to a peripheral or central disorder because treatments differ. Repositioning manoeuvres treat and resolve most peripheral causes[8]. Whereas in central disorders these do not work, and medication is typically used to reduce symptoms, alongside investigation to address underlying sinister causes that may require urgent intervention (e.g. brain tumours)[2,10].

Furthermore, when BPPV is identified, there is a lack of evidence to guide how many treatments are needed, what the expected recurrence rates are, and why some patients do not get better.

This research fellowship aims to build on the existing literature and address these critical gaps in understanding, which could refine diagnostic frameworks, and improve treatment outcomes. This aligns with patient priorities for quicker, more accurate diagnoses and the broader need for evidence-based management of positional dizziness.

2. Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients/service users, carers, and health and care services?

Positional dizziness is a prevalent complaint in healthcare[1,3], yet it is frequently misdiagnosed[6], and can have life-threatening consequences[1,7]. Frequent delays in treatment[3,11] results in increased strain on healthcare resources[3,11] and greater impact on patient quality of life[3]. By developing a practical and evidence-based diagnostic and treatment framework, this programme of work has the potential to:

Enhance Diagnostic Accuracy: Equip clinicians with tools to differentiate peripheral from central causes of positional dizziness[10], ensuring patients receive timely and appropriate treatment.

Support Healthcare Providers: Provide clinicians with accessible, evidence-based tools to confidently diagnose and manage positional dizziness, reducing misdiagnosis[6] and unnecessary referrals[11].

Improve Quality of Life: Accurate and early diagnosis and treatment will enable patients to return to work, live independently, and fully engage with family and social activities[3]. This will significantly reduce the mental and physical burden associated with positional dizziness, such as anxiety and social isolation.

Reduce Falls and Injury Risks: For elderly patients, positional dizziness is a major risk factor for falls[4], which can lead to severe injury or death[4,5] either directly or through complications. Reducing fall risks will have a profound impact on patient safety and long-term health outcomes.

Therefore, this research not only addresses immediate clinical needs but also has the potential to transform care pathways, reduce healthcare costs, and optimise resource allocation within the NHS. By alleviating the burden of dizziness on individuals and healthcare systems alike, this work anticipates meaningful, long-term impact on public health and wellbeing.

3. Review of the existing evidence - how does the existing literature support this proposal?

What causes positional dizziness?

Positional dizziness can result from either an inner ear (peripheral) cause, or due to a condition affecting the brain (central).

The commonest cause of peripheral positional dizziness is BPPV, also the most common cause of dizziness worldwide, affecting an estimated 2.4% of the population over their lifetime and with an annual incidence of 0.6%[1]. For a UK population of 68,265,200 people[12], a total of 409,591 people per year will be affected by BPPV.

Positional dizziness can also be caused by 'structural' central disorders, such as vertebrobasilar stroke and tumours, or more commonly due to 'non-structural' causes such as vestibular migraine[10]. **A neurological examination may otherwise be normal, with the person only having position triggered nystagmus and dizziness[7].** This leads to misdiagnosis and an assumption BPPV accounts for the symptoms and could have life-threatening consequences. Therefore, improving awareness and early diagnosis of central positional dizziness is critical to accessing the correct treatment.

How is BPPV diagnosed?

BPPV can affect any of the three semicircular canals in either inner ear, with the majority affecting the posterior canal (85-95% of cases)[8]. There are five main subtypes of BPPV, each eliciting specific nystagmus features in the tests[3,8].

Diagnostic criteria for BPPV[8] exist, emphasizing the importance of observing eye movements (nystagmus) characteristics during canal-specific positioning manoeuvres, in order to make a diagnosis.

How many BPPV treatments are needed, and does it recur?

Resolution rates for the most common subtype of BPPV are 72-74% after one treatment, increasing to 90% after two treatments, depending on the manoeuvre used[13]. **However, limited evidence exists regarding the optimal number of treatments required for each of the five main BPPV subtypes[3,8].**

Indeed, in a survey I conducted in 2024 among 24 multidisciplinary professionals attending a UK dizziness conference, there was no consensus on the appropriate number of treatments needed for resolution of the five main subtypes of BPPV. This knowledge gap results in patients not receiving evidence-based care, and other causes not being considered.

Additionally, BPPV is reported to recur at a rate of 36%[13]. While cardiovascular and metabolic risk factors have been proposed[14], inconsistencies in findings and methodological issues limit the evidence.

In work package 2, I will study 250 participants (see sample size section) with positional dizziness, over 18

months to develop evidence on the number of treatments required and recurrence rates for the five main subtypes, in order to guide clinical practice.

What are the challenges in diagnosing central causes?

Central causes are suspected when atypical patterns of nystagmus are seen in the diagnostic manoeuvres[10]. Yet, emerging research now suggests there may be atypical forms of BPPV that may account for these nystagmus patterns[15], complicating the process of diagnosis[3,8], and increasing the risk that central causes may be overlooked and not appropriately managed[6].

Whilst specific nystagmus features may help identify central positional dizziness, at present central pathology is often only considered when BPPV treatments fail[10]. **However, given uncertainty over the number of treatments required to resolve the five main subtypes of BPPV, relying on "failure to respond" may contribute to missed or delayed diagnosis, ongoing symptoms, and exposes patients to unnecessary treatment manoeuvres.** Once a central cause such as vestibular migraine is identified, this can then be treated appropriately[16].

In work packages 1 and 2, I will review and expand on the published literature and identify the diagnostic characteristics for central positional dizziness, to improve early detection and treatment of central causes.

Why do some patients not get better?

Effective treatments for BPPV have level 1 evidence supporting their efficacy[3,8]. These manoeuvres aim to reposition displaced debris from the affected semicircular canal by moving the patient's head and body through specific positions. Each position leverages gravity to clear debris from the canal, assuming a standard canal orientation across all individuals.

Mathematical modelling indicates there are anatomical variations in canal geometry, which may explain why approximately 10% of patients do not respond to treatment[17]. **However, no imaging studies have investigated if there is a variation that could account for treatment non-response.**

This unexplored area represents a critical gap in our understanding of BPPV management and those that do not respond. Thus, in work package 3, a subgroup of 50 participants (see sample size section) will have exploratory 3-dimensional models created from MRI scans of the inner ear. Identifying an anatomical variation could inform future studies and refine the treatment manoeuvre positions to improve treatment outcome, reduce falls risk and mortality[4,5]. These were research priorities agreed by the patient focus group, to improve the speed and accuracy of diagnosis and treatment.

4. What are the research question/aims and objectives?

Research aims:

- Review (i) the key diagnostic characteristics for central causes of positional dizziness and (ii) explore which differentiate central from peripheral dizziness (work package 1)
- Characterise the diagnostic features for peripheral and central causes of positional dizziness (work package 2)
- Determine how many treatments are required and recurrence rates for each subtype of BPPV (work package 2)
- Explore possible anatomical differences to the inner ear affecting treatment response (work package 3)

5. Project Plan

This fellowship will be formed of three interconnected work packages.

Work package 1: Systematic review

Objectives:

- Systematically review the nystagmus characteristics for central causes of positional dizziness
- Systematically review the patient risk factors and associated symptoms for central causes of positional dizziness
- Explore what characteristics, risk factors and associated symptoms distinguish central from peripheral

causes of positional dizziness

In an existing systematic review on central positional dizziness[18], only structural disorders were included, and the included literature was restricted to three years. Non-structural causes were omitted yet it is recognised that 90% of people in a vestibular migraine attack could have positional dizziness[10]. Therefore, using PICO framework, I will undertake a comprehensive and up-to-date systematic review that includes adults with positional dizziness (P), confirmed by positional manoeuvres (I), compared to peripheral positional dizziness (C), to identify the diagnostic features and associated symptoms for central positional dizziness (O).

I will register the systematic review with PROSPERO and meet with a UCL librarian to ensure a comprehensive literature search is conducted of key databases.

The articles will be screened against predefined inclusion criteria in two stages, in collaboration with my research supervisor (Professor Kaski). Initially, the titles and abstracts will be screened and then any excluded will have the full text reviewed. Disagreements will be settled by a third reviewer, who is a member of the supervisory team with expertise in undertaking systematic reviews (Dr Ellmers).

Included articles will be assessed for quality by two reviewers to reduce any bias, with a third reviewer involved if there are any disagreements. Two reviewers will extract relevant data using a preformatted data extraction form. This will encompass the inclusion criteria (PICO) and specific details of the results. Risk of bias will be assessed using ROBINS-E. The results will be reported using standardised guidelines, such as the PRISMA checklist. I will seek support, as required, from my supervisory team who have experience in systematic reviews.

The diagnostic features identified from this systematic review will be embedded in the case record form used for work package 2 and contribute towards addressing the second aim of this research fellowship.

Work package 2: Observational cohort study

Objectives:

- Characterise the nystagmus features for peripheral and central causes of positional dizziness
- Establish accompanying symptoms and patient demographics that distinguish peripheral and central causes of positional dizziness
- Determine how many treatments are required for each subtype of BPPV
- Ascertain the recurrence rates for each subtype of BPPV

This work package will prospectively assess all patients presenting with dizziness that are triggered by head and body position changes. All patients who present with BPPV will be treated, and those that do not fulfil the diagnostic criteria for BPPV or present with abnormalities in the eye examination referred for further review by a specialist Neuro-otologist. It will expand on the systematic review to establish diagnostic differences between peripheral and central causes of positional dizziness. It will also address critical gaps in the published literature on the number of treatments required and recurrence rates for each subtype of BPPV. The overall aim is to improve accurate assessment and early treatment for patients with positional dizziness, whether that be due to peripheral or central causes, priorities reported by the patient focus group.

Population

The North Central London (NCL) area is home to a diverse range of ethnicities, which includes white, Asian, and Black people. The largest ethnic minority communities in this area of London are Turkish, Irish, Polish, and Asian. NCL is the second most deprived integrated care system in London[19]. These factors have been accounted for in the development of this study, so that funding is available for participants travel and the provision of interpreters.

The study will be conducted at a single site, based at The National Hospital for Neurology and Neurosurgery. This will ensure access to the specialist equipment (Rotundum chair) and acknowledges recommendations for future research in positional dizziness[10]. The chair will allow a measurable (to 1 degree) and standardised position for each patient undergoing the assessment and treatment manoeuvres, whilst simultaneously recording eye movements[20]. These chairs are equally effective to bedside manaourvres for treatment [21] but allow more detailed analysis of eye movements, to help identify the distinguishing features between peripheral and central causes of positional dizziness and address this research fellowships aims. This will better inform bedside manoeuvres in the majority of settings in which mechanical chairs are not available.

The study will be embedded within the Acute vertigo service, and Same Day Emergency Care service (SDEC), at University College London Hospitals (UCLH) with oversight by a Consultant Neurologist (DK). This would be the only location patients from NCL would receive acute neurology care within the emergency department.

This prospective study will be open for recruitment for 18 months. Purposive sampling from this diverse population, will allow all those who present with positional dizziness to access assessment and treatment, to address the study aims. The patient advisory group will provide guidance during the study to maximise recruitment from under-represented groups to ensure the results are generalisable. Figure 1 summarises the flow of each participant through the study duration.

Figure 1: Participant flow through the study.

Sample size

Dizziness has a lifetime prevalence of 35%[1]. Among a population of 490 patients with dizziness, 20% were reported to have dizziness triggered by position changes[7], of which 80% attributed to BPPV and 20% to central causes. To ensure sufficient statistical power, a prevalence rate of 20% was used to calculate the required sample size for this study investigating positional dizziness.

The sample size was calculated with the support of statistical advisors, using the formula $n=[(z)^2 (p)(1-p)]/(d)^2$, where z is 1.96 for 95% confidence, *p* (prevalence) is 0.20 and d (margin of error) is 0.05[22]. This resulted in a required sample size of 250 participants to estimate the prevalence of positional nystagmus with a ±5% precision.

Recruitment feasibility is supported by recent data from the UCLH acute vertigo service, where 240 patients with dizziness (48 with positional dizziness) were seen in six months. Additional sources, including GP surgeries, community teams, and self-referrals, make this target achievable over 18 months.

Participant recruitment

The study will be advertised through the NCL Primary Care network and community services for North Central London. This will include approximately 200 GP surgeries, and community neurology teams in Barnet, Camden, Enfield, Haringey, and Islington. It will also be advertised in NCL emergency departments, NCL falls services and internally at UCLH by email and attendance at departmental meetings. This broad recruitment strategy will ensure the desired sample size will be met in the allocated recruitment time.

Advertisements will include screening questions for positional dizziness. These questions are adopted from a diagnostic framework for vestibular causes of dizziness[23], which was developed with patient involvement. Patients guided the recruitment approach, and reported they would like an email and a telephone option on the advertisements, both of which will be included.

Participants can be referred by their GP or community therapy services from within North Central London. Internal referrals from NCL emergency department (ED), NCL falls service, NCL hyper-acute stroke unit (HASU), and from NCL therapy services will also be accepted. In addition, participants will be able to self-refer, to help reduce the delays in diagnosis and treatment, two themes that were voiced as frustrations by the patient focus group. This broad recruitment strategy will help increase recruitment of under-served groups.

Eligibility

Adults (over 18 years old) reporting dizziness, vertigo, or unsteadiness that is triggered by head or body movements will be eligible to participate. Exclusion criteria include medical instability (cardiovascular, cervical, or neurological), spontaneous nystagmus in sitting, pregnancy, and previous history of existing and persistent vertigo. Consultee consent will be accepted for those unable to consent.

Screening and consent

Potential participants will be sent the participant information sheet electronically, as guided by the patient advisory group. They will be given the opportunity to ask questions on the phone or in person, prior to providing consent.

Electronic or paper-based consent will be obtained and stored in a central REDCap database set up for this project. Participants will then be invited to attend the hospital for assessment at the earliest point in time and their GP informed of their participation.

Participant support

Potential participants can request the information sheet and consent form in paper format, as suggested by two members of the patient focus group who did not have access to a computer. The signed paper copy of consent will then be loaded into the secure database by the research team.

Potential participants who do not speak adequate English, will be able to either ask a family member to translate, or request an interpreter is provided for them throughout this study. Access to an interpreter was included on request of the patient focus group. Visual aids and a clear explanation of the manoeuvres and tests will be given to each interpreter to translate.

To maximise participation of under-served groups[24], participants will be able to select the time they visit the hospital and travel costs will be remunerated promptly after each visit. Any non-attender will be contacted to ensure any barriers to participation are mitigated, with another appointment arranged if they wish to continue. However, participants have the right to withdraw from the study at anytime without adversely affecting their ongoing NHS care.

Data Management

Participants' data will be identifiable for the purpose of follow up, and scanning, but will be anonymised for any dissemination. All data will be kept confidential and securely stored electronically, requiring a password to access which will only be known by the research team. Data will be handled securely in line with UCLH routine data governance regulations.

Assessment visit

Participants will complete routine validated questionnaires used in dizziness and balance services (i.e. DHI, SVQ, STAI-form Y, and VIMSSQ-short) and provide more personal information relating to their health. This will help identify any accompanying or distinguishing features of peripheral and central causes of positional dizziness. Protected characteristics will also be collected to check for equality and inclusivity of those being recruited. These will be reviewed monthly and discussed with the patient advisory group and The NCL Community Neuroscience project lead, to help address any gaps in participation. This may include adjusting how the study is advertised or visiting specific clinics or teams in certain London boroughs.

Each assessment visit will take no more than one hour. It will include a lying and standing blood pressure check for any postural hypotension, as the commonest non-vestibular cause of dizziness with position changes[2] They will then have a full eye movement examination in sitting using the video goggles. Positional tests[3,8] using the Rotundum chair will then be completed, whilst wearing the video goggles, to allow detailed analysis of the nystagmus patterns to address the aims of work package 1 and 2.

Should BPPV be identified as per the diagnostic criteria[8], then the appropriate and established treatment manoeuvre will be provided once, using the Rotundum chair. The treatments aim to relocate the displaced otoconial debris from the affected semicircular canal, by moving the patient's head and body through specific positions. Each of these manoeuvres take no more than 3 minutes to complete.

Participants that do not fit the diagnostic criteria for the five main subtypes of BPPV in the positional tests[8] or present with abnormalities in the oculomotor examination, will be referred to the acute vertigo service for further assessment and management by a specialist Neuro-otologist (Professor Kaski). This will end the participants involvement in the study.

Follow up visits

To ensure the problem has fully resolved, participants will be followed up within two-weekly intervals, and re-treated once per visit, if required. They will have a maximum of three follow up visits and treatments.

In this time, if no nystagmus is seen in the assessment manoeuvres, i.e. they no longer have BPPV, this will end the participants involvement in the study.

Any participant that still has nystagmus, will be referred to the acute vertigo service for review by a Consultant Neurologist. This is informed by the preparatory survey of specialists who reported that if BPPV did not resolve after three treatments, they would investigate for other causes.

Recurrence

Any participant can contact the research team, at any point during the recruitment phase of the study, should they

feel their symptoms have recurred. This was requested at the patient focus group to speed up access to specialist care. They will then follow the same intervention pathway as new participants (see Figure 1).

Statistical analysis

Descriptive analyses will summarise categorical data as frequencies and percentages, and continuous data as means (±SD), or medians (IQR) based on distribution.

Regression models will be used to explore associations between clinical features and diagnoses of peripheral or central positional nystagmus/dizziness. The primary outcome will be the diagnosis (peripheral vs. central causes). Predictors include demographic variables, nystagmus features, and accompanying symptoms collected from questionnaires. The regression analysis will adjust for covariates identified in WP1, and missing data will be addressed using imputation methods. The supervisory team will ensure accurate statistical analysis.

The overall aim of this work package is to address priorities reported by the patient focus group and improve accurate assessment and early treatment for patients with positional dizziness, whether that be due to peripheral or central causes.

Work package 3: Pilot study

Objectives:

- Reconstruct 3D models of the inner ear from 3-tesla magnetic resonance imaging (MRI) scans
- Compare 3D models of the inner ear, of those that respond and do not respond to BPPV treatment, to explore if there are any anatomical variations
- Recommend adjustments to the treatment manoeuvres for BPPV resolution

Method

This work package will use a subgroup of participants from work package 2 to explore if an anatomical difference may account for why some patients with BPPV do not get better. There will be two groups of participants; those that do not respond to three treatments, and those that do. Stratified purposive sampling will be used, with the groups protected characteristics matched in order to ensure equality and inclusivity. Interpreters will be available, and travel will be remunerated, as recommended by the patient focus group.

Participants will have high resolution MRI scans of the inner ear undertaken by the UCL imaging department. I will collaborate closely with an academic radiologist (Dr Bray), leveraging his expertise to guide me in mastering the skills to use specialist software to reconstruct 3D models of the inner ear from the scans, to enable comprehensive visualisation and analysis of anatomy and fluid viscosity. Between group comparisons will be made to detect any anatomical differences that could account for lack of resolution of BPPV.

Sample size

Among the 250 participants with positional dizziness in our study, it is anticipated that 10% (25 participants) will not respond to the treatment manouevres[17]. This exploratory work package will compare these 25 non-responders to an age- and gender-matched group of 25 responders from the same population. The exploratory nature of this work aims to investigate potential anatomical differences between responders and non-responders.

Should anatomical differences be identified, the findings will inform future power calculations for larger studies to investigate this more comprehensively. This knowledge may guide modifications to treatment protocols to improve response rates in clinical practice.

Statistical analysis

High-resolution MRI scans will be used to create 3D reconstructions of the inner ear. Responders and non-responders will be compared for differences in canal angle, orientation, width, and endolymph viscosity using computational methods.

Statistical tests (e.g., t-tests) and multivariate models will adjust for age and gender to identify anatomical predictors of treatment response. Findings will inform hypotheses and future research.

6. Dissemination, outputs, and anticipated impact

Development of diagnostic criteria

Key diagnostic features for central positional dizziness identified in this project will be presented locally to the NCL primary care network, and nationally to the British Society of Neuro-Otology, for consultation. The findings will then be shared internationally to the Barany Society in order to propose diagnostic criteria for central positional dizziness and refine treatment recommendations for BPPV that doctors and physiotherapists can implement.

Publications

I will submit the results to relevant peer reviewed journals for publication, anticipating at least three papers to include:

- a systematic review to identify the key diagnostic characteristics for central causes of positional dizziness
- describing the nystagmus and any key distinguishing features between peripheral and central positional nystagmus
- presenting the findings from the first 3D reconstructive imaging study of the inner ear

Conferences

I will submit these results to specialist conferences attended by doctors and members of the multidisciplinary team, including the renowned International Barany Society, European Academy of Neurologists, and the British Society of Neuro-Otology.

I will also share my results through the physiotherapy interest group (ACPIVR), at annual national or regional training events and through their quarterly journal.

Multidisciplinary education

I will embed the findings in any education I provide through national and international courses, and through university master's programs. I will also offer to present the findings at any local departmental meetings.

Social media

I will share the results through a range of social media platforms, including the research unit at UCL, ACPIVR, and my own personal accounts.

Patients and public

I have established a patient advisory group of eight members with experience of positional dizziness, aged 31 to 75 years, representing diverse ethnic backgrounds, 5 females and 3 males. They will guide how best to share the findings to ensure the results are available for the people it affects and improves care in the future, for example:

- develop an infographic of the results.
- write layperson summaries for the professional routes of dissemination.
- present the results at a workshop for participants and members of the public to attend.
- write in the Ménière's Disease society newsletter.

Impact of the outputs:

The overall aim of this research fellowship is to simplify and improve the diagnosis and treatment of positional dizziness, to improve patient care and quality of life, and reduce the burden on the NHS services.

Additionally, should an anatomical difference be identified in work package 3, this could inform future larger scale research studies, leading to modifications to treatment manoeuvres and improved treatment outcome.

7. Project management

In order to successfully complete this research fellowship, I have put the following strategies in place:

I will hold weekly in-person meetings with my primary supervisor (Professor Kaski) to discuss progress, plan actions, and address questions. Monthly in-person lab meetings at UCL with researchers and PhD students will provide networking opportunities, peer support, and assistance with recruitment. Monthly online meetings with my full supervisory team will ensure everyone is updated and allow for addressing questions, with additional individual meetings as needed for work packages 1 and 3. Lastly, I will arrange quarterly online meetings with the patient advisory group to refine participant resources, support recruitment, and contribute to dissemination efforts.

8. Ethics

Once funding is approved, I will apply for internal ethical approval through the Joint Research Office at UCLH. I will also complete the Integrated Research Application System (IRAS) form and apply for NHS ethical approval. In order to reduce any delays, I will start these applications whilst waiting for funding approval.

9. Success criteria

Successful completion of this research fellowship will have the following criteria:

- Successfully meet recruitment target
- Determine the number of treatments for five main subtypes of BPPV
- Identify the recurrence rate per year, for each subtype of BPPV
- Confirm if an anatomical variation exists in those that do not respond to treatment
- Disseminate the findings (as per above)
- Attain a PhD
- Advance my clinical and research skills through completion of the proposed training and development plan
- Contribute to diagnostic criteria for central positional dizziness

To mitigate any risk of not achieving the recruitment target, I will monitor participant numbers each month using a target vs. actual recruitment graph. I will collaborate closely with the patient advisory group, NCL Community Neuroscience project lead and my supervisory team to make timely adjustments to recruitment strategies, ensuring that the target is met within the proposed timeframe.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

Patients and the public have played a vital role in shaping this fellowship proposal. The study question and design stem from recurring frustrations expressed by patients during routine clinical practice.

Subsequently, an online focus group of 10 patients, with a history of positional dizziness, shared their experiences. Participants were from across the UK, due to University College London Hospital (UCLH) Trust's tertiary service reach, aged 26 to 78, representing diverse ethnicities, 6 females and 4 males. Six key themes emerged:

1. **Delays in diagnosis** – Patients often encounter delays, with dizziness in bed frequently misattributed to BPPV without testing.
2. **Challenges in treatment** – A wide range of management approaches lead to inconsistent and sometimes inadequate care.
3. **Impact on quality of life** – Positional dizziness causes significant frustration, anxiety, and disruption to daily life.
4. **Awareness gaps** – Friends, family, and healthcare professionals often lack awareness that vestibular migraine can mimic BPPV.
5. **Communication barriers** – Describing symptoms can be difficult, especially for patients whose first language is not English.
6. **Access to care** – Patients highlighted the need for improved access to specialist diagnosis and treatment, suggesting self-referral pathways to expedite care.

These themes highlight the importance of the proposed study, to improve the diagnosis and treatment of positional dizziness and improve patient quality of life. Consequent to this feedback, interpreters will be available, and patients can self-refer for assessment during the recruitment period.

The focus group also reviewed the plain English summary of this proposal, recommending the aims were listed after the background to the research section, for ease of readability. This was incorporated in the plain English summary for this application.

Additionally, feedback from doctors managing BPPV in primary and emergency care has been instrumental. During annual training events, they consistently report challenges due to limited time, insufficient training, and lack of awareness about the varied causes of positional dizziness.

These insights underline the urgency of this research. This study aims to simplify the diagnosis of peripheral versus central causes of positional dizziness, reduce delays and improve access to effective treatment, ultimately enhancing patient outcomes and easing the burden on healthcare services.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

A **patient advisory group**, comprising eight members with experience of positional dizziness, aged 31 to 75 years, and representative of the study population (white, Asian, Black, and Turkish), was established through the Ménière's Disease Society, and will provide support throughout this study. Four online meetings a year will be held, to reduce travel and time burdens. All members report access to an electronic device to take part and will be remunerated for their time.

Roles and Contributions

1. Developing Resources

The group will review study advertisements and participant information sheets, ensuring they are accessible, free from jargon, and culturally appropriate. They have a critical role to ensure potential participants understand the study's purpose and feel comfortable engaging with it.

2. Recruitment

The group will help develop community-specific recruitment strategies to include wording in advertisements, to maximise participation. This will occur prior to study commencement and at quarterly meetings when a summary of the protected characteristics will be presented. They will help address gaps in participation, ensuring underrepresented populations are included.

3. Dissemination

The group will collaborate to create dissemination materials to maximise the study's impact. This includes preparing plain English summaries for peer-reviewed journals, contributing to newsletters, developing an infographic, social media posts, and public presentations.

Training and Support

To empower the advisory group to contribute effectively, training and support will be provided:

- **Research design:** Members will receive an overview of the study's objectives, methodologies, and expected outcomes, ensuring they feel confident in their roles.
- **Resource evaluation:** Online training by the NIHR Learning for Involvement will cover best practices for evaluating participant-facing materials, focusing on accessibility and inclusivity.
- **Dissemination:** Members will receive guidance from the NIHR Learning for Involvement, NIHR Evidence and Plain English Campaign's websites on creating impactful plain English summaries, preparing presentation materials, and using digital platforms such as social media.

In addition, throughout the project support will be offered, including time to address questions or challenges.

Impact and reporting

The public involvement in research impact toolkit (PIRIT) will be completed in collaboration with the patient advisory group. This will help to plan, track and evaluate public involvement.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Not applicable.

8. Training & Development and Research Support

Proposed training and development programme

Proposed formal study

This PhD will be registered at UCL, and I will attend free workshops provided as part of their doctoral skills training program. These include research governance and copyright, public engagement and impact, data visualisation, thesis writing and advancing my presentation skills. These sessions will be distributed across the three-year period, as outlined on the GANTT chart.

Academic training plan

To enhance these workshops, I have identified external courses that will help me to gain the skills required to conduct the proposed research study (see also GANTT chart).

1. **Gain skills in conducting a systematic review and meta-analysis**

In order to complete work package 1, I will attend a four-day course "*Introduction to systematic review and meta-analysis*" provided by the University of Bristol (£880.00). This course will teach the methodology of a systematic review, how to perform a comprehensive literature review, how to evaluate the quality of a systematic review, and how to summarise the findings of a systematic review and meta-analysis. I will also seek support from my supervisory team, which includes a researcher with experience in systematic reviews and meta-analyses (Dr Ellmers).

2. **Develop skills in Epidemiology**

In work package 2, I will collect a wide range of data, from questionnaires and clinical assessment in order to build predictive models. I will therefore attend a 15-day in-person course "*Introductory Course in Epidemiology and Medical Statistics*", provided by London School of Hygiene and Tropical Medicine (£3700.00). This course will teach me the concepts of epidemiology, statistical analysis and regression, data analysis and interpretation.

3. **Develop skills in data analysis**

I envisage collecting a large range of clinical data in work package 2, which I have not handled in previous research. Therefore, I will attend the four-day course "*Practical statistics for medical research*" provided by UCL (£750.00). This will provide the knowledge and skills to select, execute and interpret the required data analysis.

In addition, I will attend the two-day course "*Data analytics with Python: Introduction*" provided by City Lit (£249.00). This will help me to learn how to use coding to organise, examine and analyse data, which will be a useful skill when handling large datasets.

4. **Gain skills in assessing and treating positional dizziness**

I will increase my experience in using a specialist chair (Rotundum chair) that standardises the diagnosis and treatment of positional dizziness, alongside developing skills in interpreting the nystagmus collected by the video goggles. These will be essential skills to develop for work package 2, as this will be the main data collection method.

I have approached Dr Dan Dupont Hougaard, at Aalborg University Hospital in Denmark. He is an internationally renowned ENT specialist with expertise in using mechanical repositioning chairs for over 10 years and is the most published author on the topic. He has invited me to spend time within his clinic to gain skills in using the Rotundum chair and in collecting and interpreting nystagmus recorded by video-nystagmography.

Following this overseas learning opportunity, I will join monthly clinics led by audiology and neurology colleagues in the acute vertigo team at UCLH. This will provide additional opportunities to observe experts assessing patients with positional dizziness and provide practice in interpreting the nystagmus captured by video-nystagmography. I will collaborate with them to verify my analysis of the nystagmus recordings, enhancing my accuracy and ensuring the data collection methods are robust.

5. **3D modelling**

In order to complete work package 3, I need to learn skills in using specialist software to reconstruct 3D models to make comparisons. Therefore, I will attend training provided by an academic radiologist (Dr Bray) and complete online training to gain these skills. I will also collaborate with Dr Bray to verify my initial analyses are accurate.

6. **Acquire skills to maximise knowledge translation**

To make a meaningful difference to the management of positional dizziness, I need to ensure the findings from this fellowship, and future studies I design as the next part of my NIHR academic pathway, are successfully implemented. Therefore, towards the end of this fellowship, I will attend a six-week, part time course on "*Introduction to Implementation science*" provided by Kings College London (£1400.00). This course will provide skills to help with dissemination, engage stakeholders and design future studies.

Additionally, I wish to include patients and public in disseminating the research findings, to ensure they are

communicated effectively, in understandable language and reach those that are affected. I will therefore attend a free, four-week, part-time course on "*Public Involvement in Research"* provided by Imperial College London. This course will ensure I maximise the skills of the patient advisory group and other collaborators for dissemination.

7. **Develop confident skills in research leadership**

The fellowship will be crucial to allow me to complete a PhD and progress my clinical academic career. I am keen to build on my experience in clinical leadership, with a view to lead a team of researchers to continue investigating positional dizziness and expand on the findings from this proposal. I will therefore attend the free "*Resilient Leadership in Action*" course provided by UCL. This course is designed for early-stage career researchers and will help me understand my leadership style, help to set up an effective research team and equip me with the skills to lead that research team to maximise research impact.

Career development training

I will undertake this research fellowship full-time. Work package 2 will advance my skills in assessment and treatment of the most common cause of dizziness and enhance my ability to triage and recognise central disorders. These activities will be invaluable in pursuing my career as an academic researcher and developing clinically relevant research proposals in the future.

Additionally, I will observe clinics once per month with the acute vertigo team. These activities will ensure breadth to my clinical exposure and will also assist with patient recruitment.

I will also maintain my strong connections within special interest groups within physiotherapy (ACPIVR and ACPIN) and the medical community (BSNO and Barany) through attendance at yearly conferences and quarterly seminars.

RESEARCH SUPPORT / SUPERVISOR 1

Name	Professor Diego Kaski

Institution	SENSE Research Unit, Department of Clinical and Movement Neurosciences, Institute of Neurology

Position	Consultant Neurologist and Honorary Associate Professor

Description of support provided
Professor Diego Kaski is the Director of the SENSE Research Unit within the Department of Clinical and Movement Neurosciences at the Institute of Neurology. His work focuses on the translational research to develop novel therapies for patients with neurological balance disorders. His research programme aims to improve clinical outcomes in patients with vestibular and neurological balance disorders, focusing on novel clinical biomarkers to monitor disease progression and treatment response, and design and implementation of treatment strategies. He is one of the only Neurologists in the UK with expertise in central positional dizziness and is therefore strategically placed to oversee this fellowship project. He will be providing research support focusing on study design, recruitment strategy, research ethics, and ensuring successful completion of the study. He will also be supporting the development of Amanda's research, teaching and clinical skills, particularly with respect to career satisfaction, career management and progression. His background as a clinical-academic neurologist, successful research trajectory alongside his clinical expertise, and established track record in working with numerous undergraduate and postgraduate students to completion makes him an ideal supervisor for Amanda. He has overseen successful completion of 2 PhD students, and currently has 4 PhD students as primary and 2 as secondary supervisor. Professor Kaski holds monthly lab meetings, which Amanda will attend. He will also be holding 1:1 semi-structured meetings with the applicant throughout the Fellowship and will be available most days to discuss any concerns or issues as required.

RESEARCH SUPPORT / SUPERVISOR 2

Name	Dr Gita Ramdharry

Institution	National Hospital for Neurology and Neurosurgery

Position	Consultant Allied Health Professional in Neuromuscular Diseases

Description of support provided
Dr Gita Ramdharry is a Consultant Allied Health Professional specialising in neuromuscular diseases at the National Hospital for Neurology and Neurosurgery, UCLH, having begun her career as a clinical physiotherapist. She has been affiliated with UCLH for over 20 years. She is the principal investigator of the Neuromuscular Rehabilitation Research Group and has a special interest in balance in these cohorts. Dr Ramdharry possesses significant research and statistics experience and holds academic appointments as an Honorary Associate Professor at UCL. She has a strong understanding of the NIHR landscape, particularly the Integrated Clinical and Practitioner Academic (ICA) programmes and is currently serving as a Deputy Chair of the ICA Clinical Doctoral Awards panel. Her considerable supervisory experience includes mentoring NIHR ICA Clinical Doctoral Fellows and 5 PhD students to completion. Dr Ramdharry's inspiring combination of physiotherapy, NHS, academic, NIHR and mentoring experience put her in an extremely strong position to support me in both my career development and in facilitating collaboration with UCLH. Furthermore, I believe her extensive professional networks, which span neurology-focused and allied health groups, along with her active involvement in national and international committees, may offer opportunities to establish collaborations with a broad range of relevant professionals. Of significance, Dr Ramdharry has a strong commitment to equality, diversity, inclusion (EDI), and patient advocacy. Gaining her insights and guidance in these key areas will empower me to adopt best practices, ensuring that my current, intended ACAF, and future research remains inclusive, equitable, and patient-centred.

RESEARCH SUPPORT / SUPERVISOR 3

Name	Dr Toby Ellmers

Institution	Department of Brain Sciences, Imperial College London

Position	Sir Henry Wellcome Research Fellow

Description of support provided
Dr Toby Ellmers is a Sir Henry Wellcome Research Fellow in the Department of Brain Sciences, Imperial College London, where he studies the neuro-cognitive factors contributing to dizziness and imbalance in older adults. In 2023, he received the Promising Scientist Award from the International Society of Posture & Gait Research for his work. Dr Ellmers is the co-lead of the 'Concerns about Falling' Working Group within the World Falls Guidelines global initiative. This involved conducting a modified Delphi with ~100 of world experts within falls prevention and management. As part of this process, Dr Ellmers led a systematic review and meta-analysis exploring if concerns about falling predict future falls in older adults (manuscript in revision at Age and Ageing). At Imperial College London, Dr Ellmers works alongside Professor Adolfo Bronstein and Dr Barry Seemungal, eminent Consultant Neuro-Otologists and vestibular scientists. He has recently led the generation of clinical guidelines for the assessment of positional dizziness in older adult fallers (https://doi.org/10.1093/ageing/afad206) and a systematic review and meta-analysis that identified dizziness as a crucial risk factor for falls in older adults (https://doi.org/10.1093/ageing/afae177). Dr Ellmers has extensive expertise in specialist statistical analysis and the research methodologies utilised in this project (e.g., systematic reviews and meta-analyses, Delphi, large clinical datasets, etc.). He will provide hands-on

research training and support to the applicant on these methodologies throughout the Fellowship. He is currently the primary supervisor for 3 PhD students and is secondary supervisor for 4 PhD students, 3 of whom are in their final year.

RESEARCH SUPPORT / SUPERVISOR 4

Name	Dr Timothy Bray

Institution	UCL Centre for Medical Imaging

Position	Academic Consultant Radiologist

Description of support provided

Dr Timothy Bray is an Academic Consultant Radiologist based between the UCL Centre for Medical Imaging (CMI), the UCL Hawkes Institute and the UCLH Imaging Department. He has an outstanding cross-disciplinary skillset including both technical developments of novel MRI methods (with a recent patent submission) as well as MRI translation and radiology. In total, he has over 40 publications including four publications in the top-ranked MRI methodology journal and multiple top-ranked publications in MRI translation and radiology and has been awarded >£3.5m grant funding for his MRI research.

His unique, interdisciplinary skillset has been recognized through election to the International Society for Magnetic Resonance in Medicine Junior Fellowship (2019) and Annual meeting Program Committee (2025-2028). He leads the UCL/UCLH Integrated Academic Training Program in Radiology, and is Clinical Lead for the UCLH MRI service,. He has supervised two PhD students to completion and currently supervises two more, as well as one masters and one iBSc student.

He is Deputy Director of the newly-formed UCL Research Imaging Organisation (RIO), which brings together world-leading infrastructure for imaging research at UCL under a single umbrella, and will help to oversee and facilitate the developments proposed in this project.

Dr Bray has agreed to supervise this project, and meet regularly throughout the fellowship to support MRI sequence setup and the development of a processing pipeline; this collaboration will also give me access to world-leading experts in medical physics and statistics at CMI and RIO, providing an outstanding environment within which to conduct my research.

Collaborations

International collaboration

This fellowship provides an opportunity to strengthen relationships with Dr Dan Dupont Hougaard, through an overseas visit and through online meetings. We will continue to explore future research collaborations relating to positional dizziness management, to build on findings from this fellowship and his current investigations. This will allow a more varied population to be studied and increased generalisability of results for patient care improvements.

North Central London (NCL) networking

I have developed a relationship with The NCL Community Neuroscience project lead, Dr Michelle Balaratnam, to understand the impact of dizziness on patients in the emergency services, primary care and community services across NCL. Their neuroscience network will be crucial for advertising the study broadly and ensuring underserved groups are included. I will also build further relationships and visit surgeries and other clinical services to promote the study, ensure equality and inclusivity in participation, and also to disseminate the findings.

Meniere's Society (UK balance and dizziness Charity)

I have established associations with the Meniere's Society and have worked alongside the Director as a member of the Balance Interest Group. More recently I have published in their magazine and recruited members of the patient advisory group through them. I will continue to develop this relationship for the dissemination streams through presentations, newsletters and their balance awareness week events upon completion.

Physiotherapy networks (ACPIVR)

I was one of the founding members of this growing special interest group, which now has over 1200 members. Despite completing my term, I remain passionate about the aims of the group and frequently provide education for members. I will use these links to ensure the results are disseminated through their local and national conferences, workshops and bi-annual journal.

UCL relationships and support

This fellowship will be conducted out of the SENSE research unit at UCL, comprised of distinguished clinical researchers, Masters and PhD students and research fellows from across the globe. I will join monthly department meetings overseen by the unit's director (DK) and also meet with other PhD students registered at UCL. This will create a rich environment for peer support, collaboration, facilitate an international network through which to disseminate findings and help to forge future collaborations.

In addition to completing the above mentioned statistics training, I will continue to collaborate with the Medical Statistics team at UCL and members of my supervisory team with statistical expertise (GR and TE), to ensure my data analysis is thorough and accurate.

Commercial relationships

This research fellowship will use a specialist repositioning chair to diagnose causes of positional dizziness and treat benign paroxysmal positional vertigo (BPPV), which was donated as part of a philanthropic gesture to the research unit. I have established a good relationship with Norbert Wick, the managing director of Balcare, who installed the Rotundum rotatory chair and provided basic training. He will support the project with maintenance, repairs, and technical assistance. There are no conflicts of interest or funding associated with this collaboration.

Interdisciplinary networks

I have an established association with the lead Occupational Therapist (Vassilios Tahtis) at King's College Hospital, London. I trained all of the occupational therapists and physiotherapists in the department, to help them identify vestibular problems. We also published a paper together to improve training in positional manoeuvres for BPPV. He is keen to help disseminate the findings of this research.

Additionally, I have been contacted by an ENT specialist in Athens (Dr George Korres) to discuss current clinical challenges with positional dizziness. We have identified several common areas, and plan to write a manuscript for publication, that will synthesise the literature on positional nystagmus and dizziness so far.

Host Organisation Support Statement
The following statement of support must be completed by the head of your lead academic department. In order to contribute to this application, this individual must be invited to do so via the 'Participants and Signatories' section of the form.

Statement of support
As the Head of the Clinical and Movement Neurosciences Department at University College London, I wish to express my unreserved support for Ms Amanda Male's NIHR doctoral fellowship application. The Department of Clinical and Movement Neurosciences is a world-leading group that brings together expertise across the clinical and basic science spectrum of the physiology and pathology of human movement and movement disorders. Ms Male will benefit from being based in the SENSE Research Unit, which is a unique group pursuing research in translating behavioural neuroscience into experimental medicine and treatments for adults with vestibular and balance disorders. Ms Male's training as a physiotherapist and research in physiotherapy approaches to manage dizziness perfectly aligns with the multidisciplinary structure of this department.

As the host organisation, we are committed to providing the necessary infrastructure and support for the successful completion of this fellowship. We recognise the importance of this fellowship in facilitating the development of highly skilled clinical academic researchers who will make significant and direct contributions to improving patient care. We value the importance of creating a research environment that is inclusive, as evidenced by the range of subspecialty lab members, gender, and ethnicity within the team. We strive to ensure we provide a supportive environment for all our lab members and collaborators, holding weekly meetings to discuss general lab matters, offering an opportunity to evaluate research integrity, and offering 1-1 meetings with those that need additional support. We expect and support all new members to abide by the UCL principles of equality, diversity, and inclusion and ensure that members receive appropriate training in this regard, prior to starting.

We will ensure that Ms Male is provided with adequate resources, including access to facilities, equipment, and training, to enable her to carry out her research project effectively. She will have access to world-leading facilities and technology in the gait laboratory at 33 Queen Square together, most notably the Rotundum chair that is integral to her project, with the highest quality training. In addition, we will provide Ms Male with doctoral mentorship and steer from an internationally recognised clinical academic, Associate Prof Diego Kaski, an expert in behavioural neurology and vestibular disorders. The Department also benefits from direct links with experienced researchers and clinicians at Queen Square and Royal Free hospitals to support her research. Ms Male shows every sign of being at the forefront of her field and we are keen for her to join our department.

The department has an agreed pathway for the development of clinical academic posts at pre- and post-doctoral level and in the medium term we will help her conceptualise and deliver a top-quality NIHR post-doctoral fellowship application to ensure she continues on a clinical academic path in the longer term.

As the host organisation, we are committed to ensuring Ms Male's success as a clinical academic researcher throughout her fellowship and beyond. We believe that this fellowship will significantly contribute to the advancement of healthcare and the development of a highly qualified clinical academic researcher.

Yours sincerely,

Professor Tony Schapira

Head of Department of Clinical and Movement Neurosciences
UCL Queen Square Institute of Neurology and National Hospital for Neurology and Neurosurgery
Queen Square
London WC1N 3BG

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Amanda Male

Employing organisation	University College London

Grade	UCL Grade 7

Starting point on scale	Spine 37

Increment Date	

Justification of Costs
This will cover the salary costs of the lead applicant undertaking this research fellowship on a full time basis. She was last working in the NHS as a band 7 vestibular physiotherapist since 2011, and this banding therefore reflects her advanced clinical skills and experience. In addition, her expert opinion continues to be sought for education and mentoring, by physiotherapists and medical professionals, at a national and international level.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	47064
Geographical Weighting (£)	5125
Other allowances (£)	
Superannuation (£)	7828
Nat. Insurance (£)	7078
Current Annual Costs (£)	**67095**

Staff Type: (Lead Applicant)-Ms Amanda Male
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: This will cover the salary costs of the lead applicant undertaking this research fellowship on a full time basis. She was last working in the NHS as a band 7 vestibular physiotherapist since 2011, and this banding therefore reflects her advanced clinical skills and experience. In addition, her expert opinion continues to be sought for education and mentoring, by physiotherapists and medical professionals, at a national and international level.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	47064	48447	49851	145362
National Insurance	7078	7286	7497	21861
Superannuation	7828	8036	8247	24111
Geographical weighting / Other allowances (if applicable)	5125	5125	5125	15375

	Year 1	Year 2	Year 3	Total Cost (£)
Total HEI	53675	55116	56577	165368
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conferences/dissemination
Justification of Cost: Year 2 - Costs requested to cover the fees (£80) of attending the two day, British Society of Neuro-Otology meeting, in October 2027, in the United Kingdom. Year 3 - Costs requested to cover the fees (£500), approximate return flights to Sweden (£300) and approximate accommodation costs for four days (£500) to attend the Barany Society Meeting in June 2028. These interdisciplinary courses bring together specialists in the United Kingdom, and then internationally, to share research that has been completed or share provisional research findings. Attendance at these courses will be essential to maintain an up to date understanding of developments in the vestibular specialism, and to develop strong relationships with other clinical and academic researchers for future collaborations.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	80	1300	1380

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel (participants)
Justification of Cost: Costs are requested to reimburse 250 participants for travel to the research unit (£20) in order to participate in the research (work package 2 & 3). Reimbursement will help ensure equality and inclusivity of recruitment for the population to be studied. On average, participants may have up to three visits each, with costs calculated to factor in train or taxis fares for the geographical area to be studied (North Central London).

	Year 1	Year 2	Year 3	Total Cost (£)
	4995	4996	5009	15000

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	4995	5076	6309	16380

Equipment
No costs have been added

Summary

	Travel (excl Conferences)	Travel (Conferences)	Equipment

Total HEI	12000	1380	0
Total NHS	0	0	0
Total Commercial	0	0	0
Total Other	0	0	0

Consumables

Description: Copyright fees for questionnaire (STAI adults)
Type of cost: HEI
Please describe fully: Copyright fees for questionnaire (STAI adults)
Justification of Cost: This will cover the license to administer costs for a key questionnaire (STAI- form Y) to use with 250 participants. This will help identify any participant characteristics that would improve the diagnosis of central positional dizziness (work package 2). This questionnaire has copyright fees owned by Mind Garden (https://www.mindgarden.com/145-state-trait-anxiety-inventory-for-adults) equating to £2.16 per person.

	Year 1	Year 2	Year 3	Total Cost (£)
	175	175	174	524

Description: VNG related equipment
Type of cost: HEI
Please describe fully: VNG related equipment
Justification of Cost: To provide the lead applicant with replacement parts (cables, foam goggle rims and plastic covers), due to wear and tear, to ensure the videonystagmography goggles can be used throughout the recruitment period (work package 2).

	Year 1	Year 2	Year 3	Total Cost (£)
	3996	3997	4007	12000

Description: Interpreter costs
Type of cost: HEI
Please describe fully: Interpreter costs
Justification of Cost: To cover costs of interpreters for any participant (work package 2 & 3) that requests language assistance in order to ensure equality and inclusivity of recruitment for the population to be studied in North Central London. This is figure recognises that 20% of the population to be studied (n=250) may be non English speaking (https://nclhealthandcare.org.uk/wp-content/uploads/2023/12/Start-Well_Summary-Document_web-version-1.pdf). This cost is calculated using the DPSI (Diploma in Public Service Interpreting) higher rate of £40 per hour.

	Year 1	Year 2	Year 3	Total Cost (£)
	768	767	769	2304

Description: 3-tesla MRI scanning costs
Type of cost: HEI
Please describe fully: 3-tesla MRI scanning costs
Justification of Cost: This will cover the total staff and departmental costs for the 3 tesla MRI scans of the inner ear for 50 participants (work package 3). This quote was provided by the imaging department that will conduct the scans and where one of the supervisory team (Dr Tim Bray) is based.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	12674	1076	13750

Description: Register of study with the UCL/UCLH Research Imaging Organisation (RIO).
Type of cost: HEI
Please describe fully: Register of study with the UCL/UCLH Research Imaging Organisation (RIO).
Justification of Cost: This will cover the costs of registering the project with the UCLH/UCL Research Imaging Organisation, as recommended by Dr Tim Bray. This department will help the lead applicant develop the skills to

interpret the scans and create the 3D models for analysis. They will also help with imaging repository at the end of the project. This will be essential for work package 3.

	Year 1	Year 2	Year 3	Total Cost (£)
	500	0	0	500

Description: Omron HBP-1320 Blood Pressure Monitor
Type of cost: HEI
Please describe fully: Omron HBP-1320 Blood Pressure Monitor
Justification of Cost: This will cover the costs of ordering a blood pressure machine and cuffs from a provider endorsed by the NHS trust where this project will be based. This will be essential to check for any postural hypotension of participants in work package 2, as this is the most common non vestibular cause of position induced dizziness.

	Year 1	Year 2	Year 3	Total Cost (£)
	360	0	0	360

Description: Info-graphic
Type of cost: HEI
Please describe fully: Info-graphic
Justification of Cost: This will cover the cost of engaging with a graphic design service to develop an infographic, in collaboration with the patient advisory group. This can then be used to share the results of the study with patients and public.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1400	1400

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	5799	17613	7426	30838

Patient and Public Involvement

Description: PPI - patient advisory group meetings
Type of cost: HEI
Please describe fully: PPI - patient advisory group meetings
Justification of Cost: This will remunerate the eight members of the patient advisory group, for an hour of their time to meet four times per year, over the three years of this research fellowship. These meetings will be online. The eight members will be sent short documents to review ahead of these meetings, to include the study advertisements, participant information sheets and consent forms, and will be asked to provide support in recruitment approaches for under represented groups. They will help develop resources for dissemination to include an infographic of the results, review plain English summaries for publications and prepare materials for the PPI workshop. These costs have been calculated using the NIHR payment guidance of £50 per hour due to them receiving tasks that require small amounts of preparation.

	Year 1	Year 2	Year 3	Total Cost (£)
	1599	1599	1602	4800

Description: PPI - workshop costs for dissemination
Type of cost: HEI
Please describe fully: PPI - workshop costs for dissemination
Justification of Cost: This will cover refreshments for 50 participants or public (£10 per person) to attend a workshop to disseminate the research findings, as part of the balance awareness week in September 2028. In addition, each participant will be reimbursed for their time (£25 per person). These costs have been calculated using the NIHR payment guidance.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1750	1750

Description: PPI - workshop costs patient advisory group
Type of cost: HEI
Please describe fully: PPI - workshop costs patient advisory group

Justification of Cost: This will cover refreshments (£10 per person) for the 8 patient advisory group members to attend a workshop to help disseminate the research findings, as part of the balance awareness week in September 2028. In addition, the eight patient advisory group members will be reimbursed for their time. These costs have been calculated using the NIHR payment guidance of £50 per hour due to them receiving tasks that require small amounts of preparation.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	480	480

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	1599	1599	3832	7030

Other Direct Costs

Description: Printing and binding of thesis
Type of cost: HEI
Please describe fully: Printing and binding of thesis
Justification of Cost: This cost will cover the printing and binding of the PhD thesis.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	100	100

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	0	0	100	100

Training And Development

Type of cost: HEI
Description: Short Courses
Item Description: PhD training costs - An Introduction to Implementation science
Justification of Cost: This will cover the cost of attending “An Introduction to Implementation science” course through Kings College London. The course will provide the skills to assist in the dissemination of findings and ensure the findings are translated into clinical practice promptly for patients to benefit from the research. It will also equip the applicant with the skills to design future studies as part of her NIHR academic pathway, that will be successfully implemented and make a meaningful change. This course is a vital part of the applicants academic training plan (work package 1-3).

	Year 1	Year 2	Year 3	Total Cost (£)
	00	0	1400	1400

Type of cost: HEI
Description: Short Courses
Item Description: PhD training costs - Introduction to systematic review and meta-analysis
Justification of Cost: This will cover the cost of attending “Introduction to systematic review and meta-analysis” provided by the University of Bristol (work package 1). It will enable the applicant to learn the skills to undertake a systematic review, which forms the first work package of this research fellowship. This course therefore forms a vital part of the applicants academic training plan.

	Year 1	Year 2	Year 3	Total Cost (£)
	880	0	0	880

Type of cost: HEI
Description: Short Courses
Item Description: PhD training costs - Practical statistics for medical research
Justification of Cost: This will cover the cost of attending “Practical statistics for medical research” provided by UCL. The course will teach the applicant how to execute and interpret data analysis, that is required for this research fellowship (work package 2 & 3). This course is a vital part of the applicants academic training plan.

	Year 1	Year 2	Year 3	Total Cost (£)

	0	750	0	750

Type of cost: HEI
Description: Overseas Research Visits
Item Description: PhD training costs - Overseas visit for training
Justification of Cost: This will cover the cost of an overseas training visit to gain the skills in using the specialist chair and video goggles, for the diagnosis and treatment of positional dizziness (work package 2). This overseas visit will be to Denmark, to spend time with the most published and experienced clinician in using the specialist chairs. Costs requested to cover approximate return flights to Denmark (£300) and approximate accommodation costs for one month (£1000). This course therefore forms a vital part of the applicants academic training plan.

	Year 1	Year 2	Year 3	Total Cost (£)
	1300	0	0	1300

Type of cost: HEI
Description: Short Courses
Item Description: PhD training costs - Python
Justification of Cost: This will cover the cost of attending “Data analytics with Python: Introduction” provided by City Lit. It will teach the applicant how to use coding to organise and analyse large data-sets, which will be collected in this research fellowship (work package 2 & 3). This course is a vital part of the applicants academic training plan.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	249	0	249

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: PhD Fees
Justification of Cost: These costs will enable the applicant to complete a full time PhD at UCL, as a result of the research fellowship.

	Year 1	Year 2	Year 3	Total Cost (£)
	6211	6211	6229	18651

Type of cost: HEI
Description: Short Courses
Item Description: PhD training costs - Introduction into Epidemiology and Med Statistics
Justification of Cost: This will cover the cost of attending an in person “Introductory Course in Epidemiology and Medical Statistics” provided by The London School of Hygiene and Tropical Medicine. This will teach the applicant the concepts of epidemiology, statistical analysis to include regression and skills in data analysis and interpretation, which will be important skills in this fellowship (work package 2). This course is a vital part of the applicants academic training plan.

	Year 1	Year 2	Year 3	Total Cost (£)
	3700	0	0	3700

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	12091	7210	7629	26930

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	75680	81773	78600	236053

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: Estates
Justification of Cost: The estates rate is calculated to include premises-related research costs, including: repairs and maintenance, basic services and utilities, rates, rents, insurance, gross buildings depreciation, existing equipment and research. The estates rate includes estates costs that are incurred by both academic departments and the cost of all support staff related to these area. These are calculated using TRAC methodology.

	Year 1	Year 2	Year 3	Total Cost (£)
	19408	19408	19408	58224

Type of cost: Other Indirect Costs
Name of Higher Education Institute: Indirect Costs
Justification of Cost: Costs are for provision of general support, e.g. health and safety, library services, stores, equipment maintenance, and the finance department. Costs will support research infrastructure including health and safety, stores, workshops, equipment maintenance, management and administration. These are calculated using TRAC methodology.

	Year 1	Year 2	Year 3	Total Cost (£)
	68672	68672	68672	206016

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	88080	88080	88080	264240

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	70464	70464	70464	211392

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	207743	211392	419135
Total Higher Education Institution Costs (100%)s	28310	0	28310
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	0	0	0
Total Research Costs Requested From Funder	236053	211392	447445

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**53675**	**55116**	**56577**	**165368**
Travel-subsistence-and-conference-fees	**3996**	**4077**	**5307**	**13380**
Equipment	**0**	**0**	**0**	**0**
Consumables	**4639**	**14091**	**5941**	**24671**
Patient-and-Public-Involvement	**1279**	**1279**	**3066**	**5624**
Other-Direct-Costs	**0**	**0**	**80**	**80**
Training And Development	**12091**	**7210**	**7629**	**26930**
HEI-Indirect-costs	**70464**	**70464**	**70464**	**211392**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**146144**	**152237**	**149064**	**447445**

NHS Support Costs
No costs have been added

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs
No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient/service user care being provided different from the usual standard treatment for the condition?
N/A

Usual Treatment Costs
No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	0

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	447445
Total NHS (Support and Excess Treatment) Costs	0
Total Funding Required	447445

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as a .csv file or Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - References_13.1.25.docx) is included as an appendix within this file.

(Figures / Tables - Figures 11.1.25.docx) is included as an appendix within this file.

(Research Timetable - Research timetable_11.1.25.docx) is included as an appendix within this file.

(SoECAT Template - Funder Export_351718_SoECAT_PPN vs CPN.xlsx) is included as an appendix within this file.

11. Participants and Signatory

Participants
New participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please see the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

You are required to supply the names and email addresses (if not already registered with this online application system) of each required participant. Everyone **newly** named in this section will be acting as a 'participant' to your application and will need to agree to be part of this application.

By confirming participation, participants are acknowledging their involvement and input into this application, and agreeing to be involved in the proposed fellowship. You must ensure all **new** participants are happy for your application to be submitted before submitting it on the online system.

It is permissible for a single individual to serve as a signatory and a participant, but they will need to be added separately under each heading.

Details of the individual who will be required to approve your application after submission should be entered in the 'Participants and Signatory' section. Please note that it is acceptable for a Head of Department to also act as a research support role. If this is the case, they will need to be included in this section for both roles.

Participants must confirm their participation on your application before you will be able to press the submit button. Please see the 'Application Submission Process Flow Diagram' in the guidance notes for further information.

Doctoral Primary Supervisor

Doctoral Primary Supervisor Participant

Title	Dr
First Name	Diego
Last Name	Kaski
Position	Consultant

Supervisor

Supervisor Participant

Title	Dr
First Name	Timothy
Last Name	Bray
Position	NIHR Clinical Lecturer

Supervisor Participant

Title	Dr
First Name	Toby
Last Name	Ellmers

Position

Supervisor Participant

Title	Dr
First Name	Gita
Last Name	Ramdharry
Position	Consultant Allied Health Professional/ Honorary Associate Professor

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mrs
First Name	Kasia
Last Name	Haremza
Position	Assistant Director of Research Development and Support (Pre-Award)

Signatory
Participants must complete their actions on your application before submission; the signatory must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

You are required to supply the name and email address (if not already registered on the ARAMIS application system) of the individuals who will be 'signing off' your application. Once their contact details have been entered below, these individuals will be invited to log in and confirm their participation in your application.

The signatory must ALSO approve the application AFTER you have selected the 'SUBMIT' option but BEFORE the application submission deadline. Please see the Submission Process Flow Diagram and Applicant Guidance Notes for further information.

The signatory approval will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatory) will be notified of this via an automated system-generated email.
NIHR will not accept any applications unless fully approved by your signatory before the 1:00 pm deadline on Thursday 16th January 2025.

Host Organisation - Heads of Department

Head of Department or Senior Manager

Title	Professor
First Name	Anthony

Last Name	Schapira
Position	

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
I declare I have no conflicts or potential conflicts of interests, to undertake this research fellowship.

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR Research Governance Framework. If you are applying for an NIHR award which comes with partnership funding from another organisation (e.g. a charity or industry partner) your information (including your NIHR Application Form and subsequent progress/annual reports) may be shared with the partner organisation as part of the review process and ongoing monitoring of the award. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

Appendices

1. References_13.1.25.docx
2. Figures 11.1.25.docx
3. Research timetable_11.1.25.docx
4. Funder Export_351718_SoECAT_PPN vs CPN.xlsx

Appendix #1: References_13.1.25.docx

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Appendix #2: Figures 11.1.25.docx

Figures:

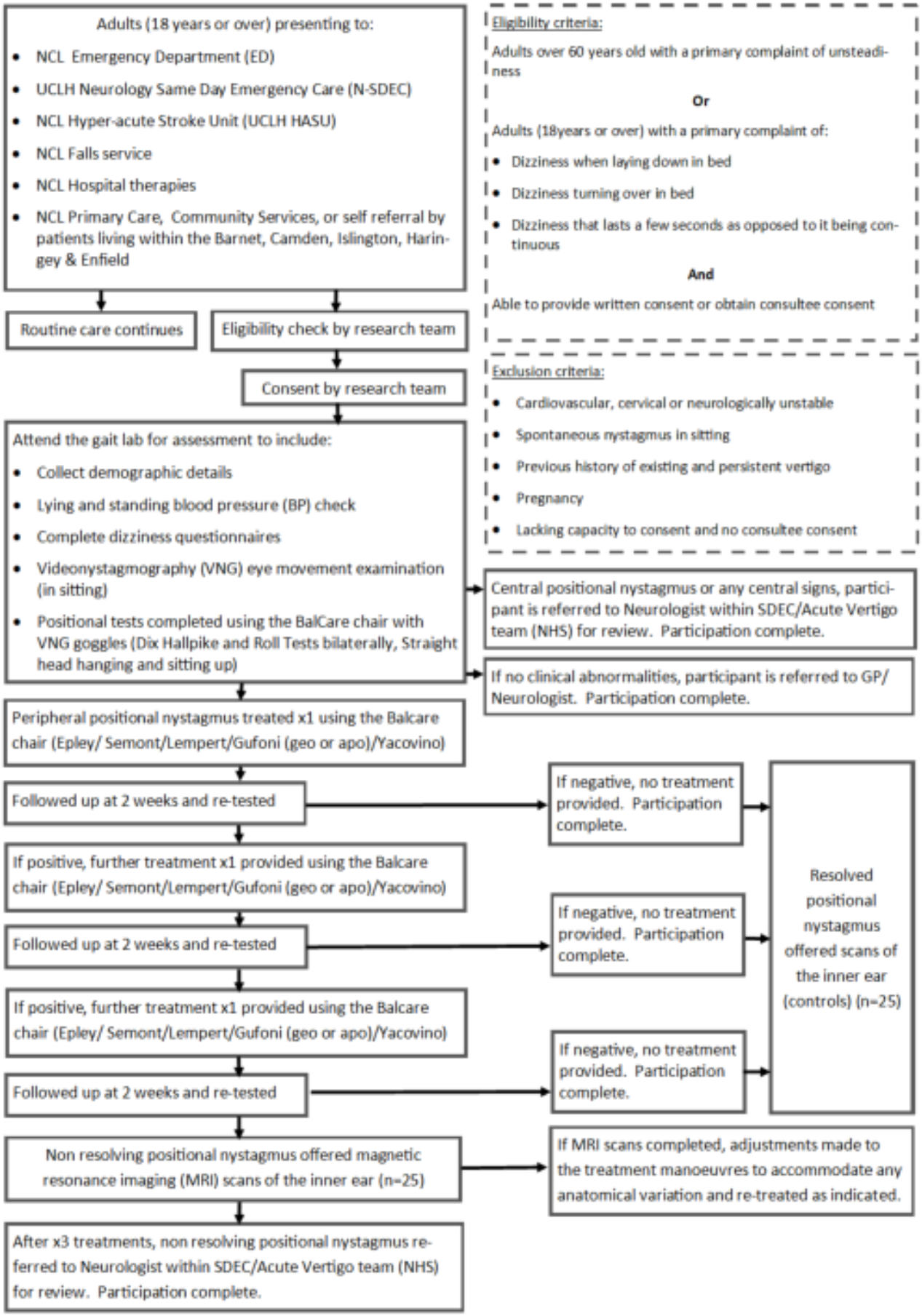
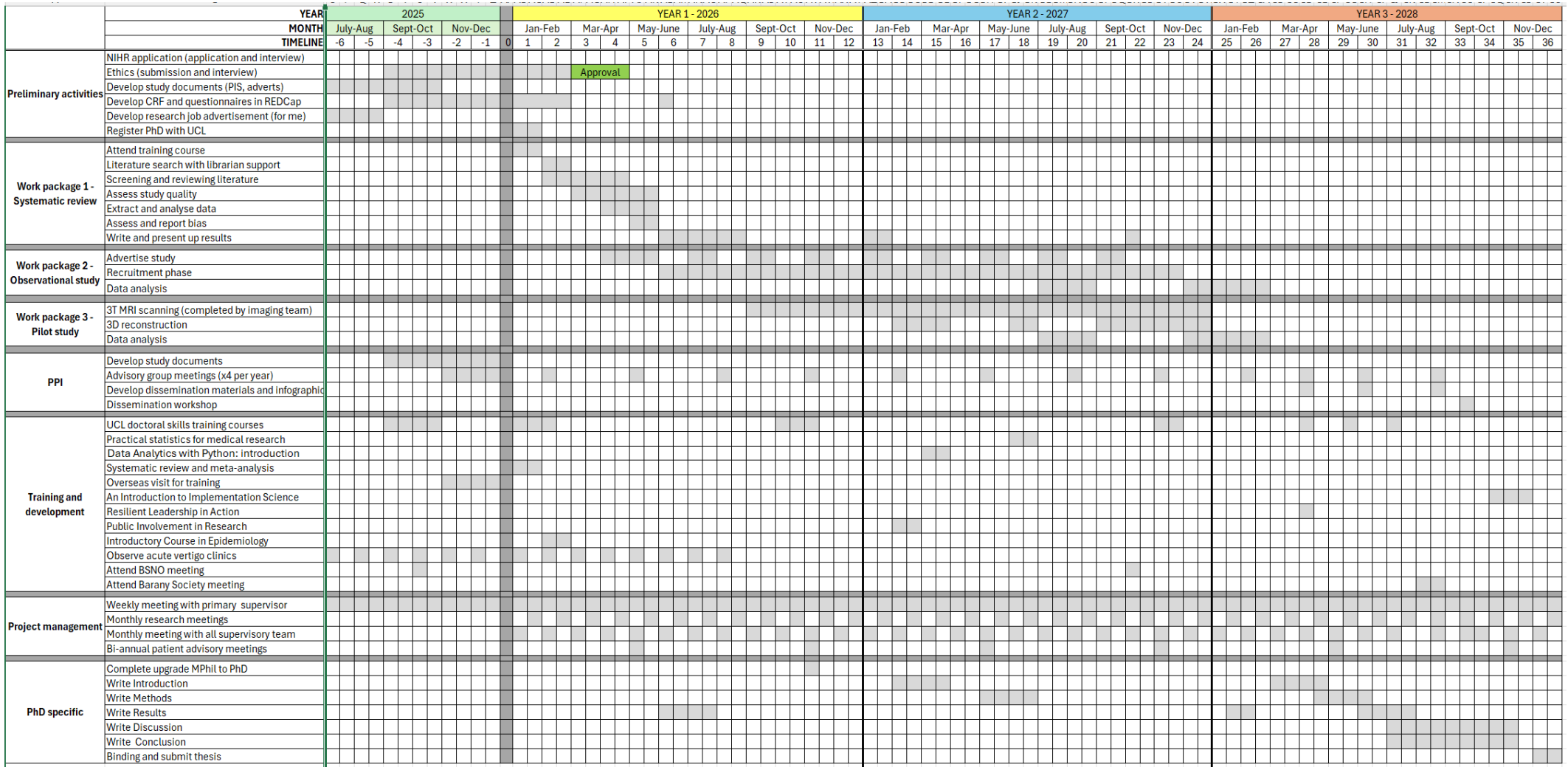

Adults (18 years or over) presenting to:
NCL Emergency Department (ED)
UCLH Neurology Same Day Emergency Care (N-SDEC)
NCL Hyper-acute Stroke Unit (UCLH HASU)
NCL Falls service
NCL Hospital therapies
NCL Primary Care, Community Services, or self referral by patients living within the Barnet, Camden, Islington, Haringey & Enfield
Routine care continues
Eligibility check by research team
Consent by research team
Eligibility criteria:
Adults over 60 years old with a primary complaint of unsteadiness
Or
Adults (18years or over) with a primary complaint of:
Dizziness when laying down in bed
Dizziness turning over in bed
Dizziness that lasts a few seconds as opposed to it being continuous
And
Able to provide written consent or obtain consultee consent
Exclusion criteria:
Cardiovascular, cervical or neurologically unstable
Spontaneous nystagmus in sitting
Previous history of existing and persistent vertigo
Pregnancy
Lacking capacity to consent and no consultee consent
Attend the gait lab for assessment to include:
Collect demographic details
Lying and standing blood pressure (BP) check
Complete dizziness questionnaires
Videonystagmography (VNG) eye movement examination (in sitting)
Positional tests completed using the BalCare chair with VNG goggles (Dix Hallpike and Roll Tests bilaterally, Straight head hanging and sitting up)
Central positional nystagmus or any central signs, participant is referred to Neurologist within SDEC/Acute Vertigo team (NHS) for review. Participation complete.
If no clinical abnormalities, participant is referred to GP/ Neurologist. Participation complete.
Peripheral positional nystagmus treated x1 using the Balcare chair (Epley/ Semont/Lempert/Gufoni (geo or apo)/Yacovino)
Followed up at 2 weeks and re-tested
If negative, no treatment provided. Participation complete.
If positive, further treatment x1 provided using the Balcare chair (Epley/ Semont/Lempert/Gufoni (geo or apo)/Yacovino)
Followed up at 2 weeks and re-tested
If negative, no treatment provided. Participation complete.
If positive, further treatment x1 provided using the Balcare chair (Epley/ Semont/Lempert/Gufoni (geo or apo)/Yacovino)
Followed up at 2 weeks and re-tested
If negative, no treatment provided. Participation complete.
Resolved positional nystagmus offered scans of the inner ear (controls) (n=25)
Non resolving positional nystagmus offered magnetic resonance imaging (MRI) scans of the inner ear (n=25)
If MRI scans completed, adjustments made to the treatment manoeuvres to accommodate any anatomical variation and re-treated as indicated.
After x3 treatments, non resolving positional nystagmus referred to Neurologist within SDEC/Acute Vertigo team (NHS) for review. Participation complete.

Figure 1: Participant flow through the study.

Appendix #3: Research timetable_11.1.25.docx

Research timetable



Appendix #4: Funder Export_351718_SoECAT_PPN vs CPN.xlsx